PACKAGING DESIGN FOR A FORMULA STUDENT VEHICLE

Formula Student FEUP 03

Department of Chassis

Department Leader:

Miguel Monteiro

Department Members Responsible:

Íris Botelho
José Conchinha
Vasco Macedo

Reviewed by:

Miguel Monteiro

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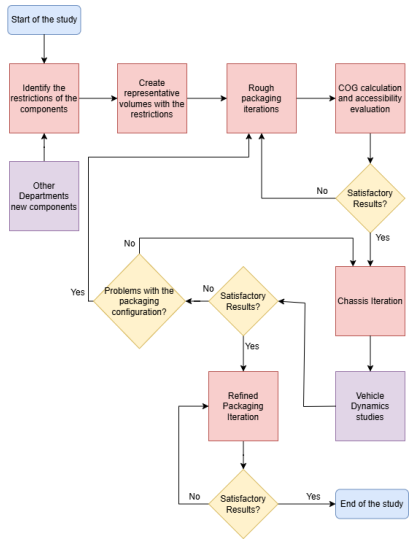


Figure 1.1: Workflow of the study and design phase

Importance of Packaging Configurations

This study can have a significant impact on the team's performance. Lowering the center of gravity offers several advantages, including improved stability, better handling, and enhanced overall dynamic behavior. It will also serve as reference parameters for the Vehicle Dynamics department; therefore, it is essential to understand the impact that the mere positioning of a component may have. Furthermore, ensuring good accessibility for key components can save the team valuable time during competitions: whether when charging the accumulator, debugging a PCB, or performing maintenance tasks. While this time saved may seem minor, it can make the difference between the team being able to participate in an event or not. Accessibility is also crucial during the manufacturing phase, as it can facilitate the work of other departments, accelerate the assembly process, and provide the team with more testing time, which almost certainly correlates with better results.

1.1.7 Variation in the car's properties

The values obtained for the height of the center of gravity are important, but they do not provide a clear interpretation of the vehicle's behavior, since the center of gravity was calculated considering only those components and using the lowest point of the accumulator as the reference.

So, to obtain a more realistic value for the center of gravity, the third packaging iteration was positioned inside the FS FEUP 02 chassis together with additional components in order to better represent the actual behavior of the system. Therefore, besides the frame, the seat, head restraint, steering wheel, driver protection elements, pedal box, and the impact attenuator were also included.

Inserir foto FSFEUP 02 com componentes

With the components positioned in the car, it becomes possible to assess the influence that the placement of each one has on the vehicle's properties. For this purpose, the positions of the components with the greatest mass and volume (those expected to have a significant impact on the car's characteristics) were varied. These components include the Master Panel, ECU, Pneumatic Box, LV Battery, Data Logger Box, Air Tanks, Accumulator, and Inverters. By varying their position along the main axes (x: lateral axis of the car, y: longitudinal axis, z: height), graphs were generated that clearly illustrate the influence of each element.

Master Panel

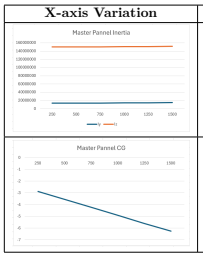
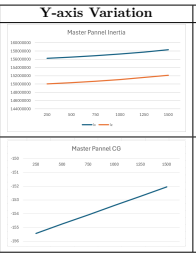
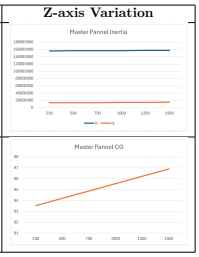
X-axis Variation	Y-axis Variation	Z-axis Variation
		

Table 1.4: Master Panel influence



Figure 1.31: Force Tab

In the 'Materials' panel, the user can choose a material from the initial dropdown banner or define custom materials. New materials can be saved permanently by clicking the 'Load new material' button.

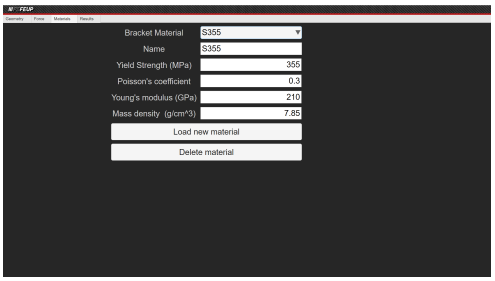


Figure 1.32: Materials Tab

Then finally, in the 'Results' tab the final stress and deflection results can be analysed. Additionally, a 3D representation can be seen to provide spatial visualization of the bracket.

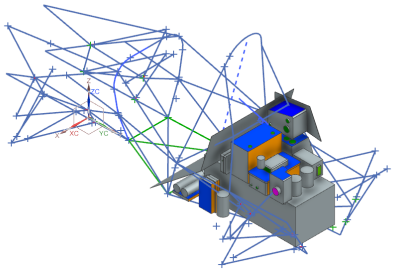


Figure 2.6: PA03_A01 [25-12-03-00]

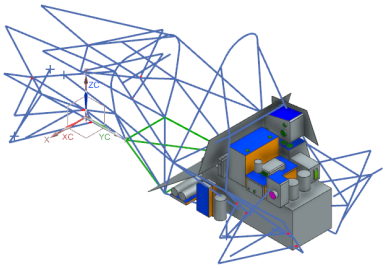


Figure 2.7: PA03_A01 [25-12-03-01]